

1. How these objects map to the demand model

In the panel nested logit, agents in a state-month market choose among $\mathcal{J} = \{\text{representative cigarette, representative e-cigarette, not smoking}\}$, and the representative e-cigarette price is

$$p_{ms} = (\text{avg. price per mL})_{ms} \times (\text{avg. e-liquid capacity})_{ms},$$

with both averages taken under **mL (e-liquid) sales weights**. That is exactly the object denoted P_{hom} . Writing u_i for units and m_i for mL per unit of product i ,

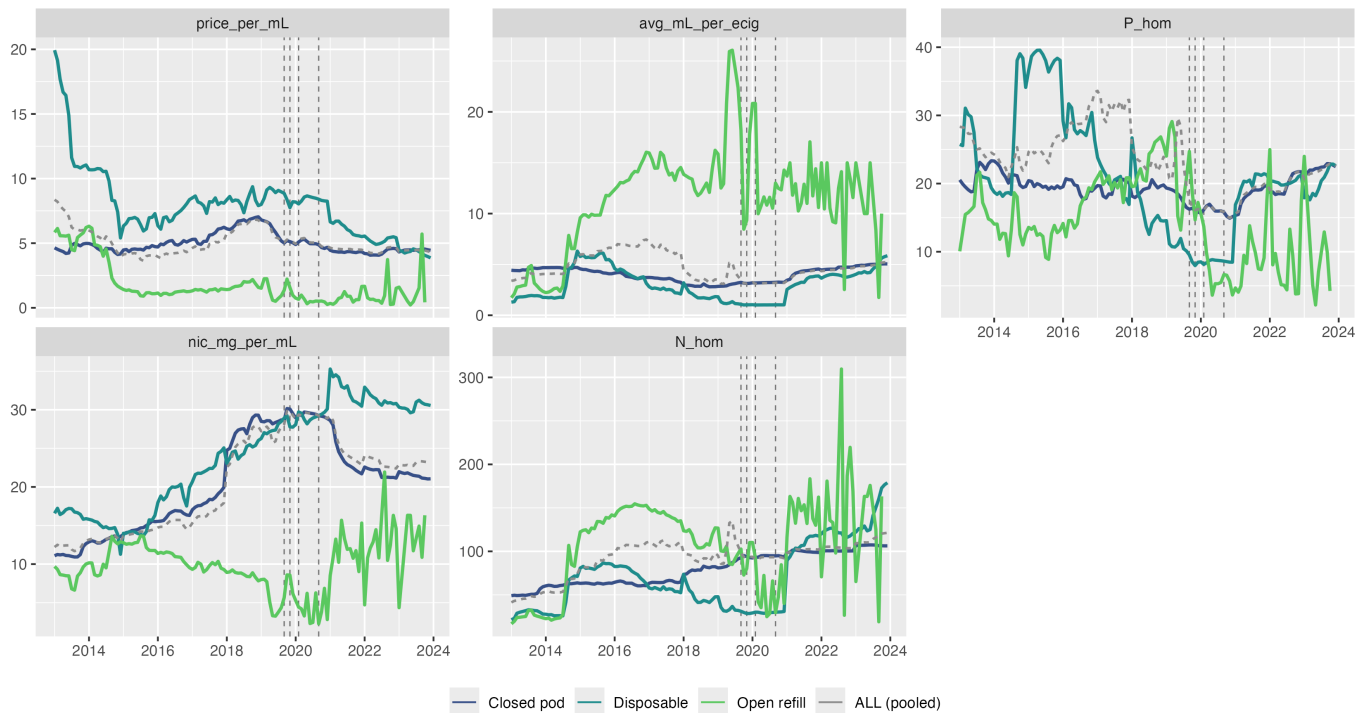
$$p_{ms} = \left(\frac{\text{revenue}}{\text{mL sold}} \right) \times \bar{m}, \quad \bar{m} = \frac{\sum_i u_i m_i^2}{\sum_i u_i m_i},$$

where $\bar{m}$ is average product size **weighted by mL sold** (the m_i^2 is not a second squaring — it is the algebra of weighting each size m_i by its own mL sales $u_i m_i$). The representative e-cigarette's nicotine content — the nicotine *flow* $n_{ij,m}$, built as $\text{mg/mL} \times \text{mL} \times \text{units} \times 0.68$ transfer efficiency and feeding the stock N_{im} — is the analogous mL-weighted object, N_{hom} .

Why this matters for the paper: the descriptive series below *are* the demand-side regressors (p_{ms} and the representative-e-cig nicotine that drives N_{im}). Anything unusual in them — the mid-2019 blip, the pod-weighting — is already inside the estimation inputs. The series are shown in §2, split by product form factor; the disposable-coverage comparison appears in §6, and the supply relation that closes the model is developed in §9.

2. What drives p_{ms} and the representative nicotine, period by period

National monthly e-cig series, broken out by T2 type (pooled = dashed grey)



Events: 1 EVALI score (fall 2019) 2 JUUL pulls fruit/mint (Nov 2019)
3 FDA flavored-cartridge enforcement (Feb 2020) 4 PMTA deadline (Sep 2020)

2013–2015 (cigalikes). blu/NJOY/Logic cigalikes and bottled juice dominate tracked retail; price per mL falls ($\sim \$8 \rightarrow \5) as bulk-juice competition intensifies; nicotine is modest, so p_{ms} is high-ish but the representative nicotine flow is low.

2016–2017 (nicotine salts). JUUL commercializes salts, making $\sim 5\%$ nicotine palatable in 0.7 mL pods. The representative concentration climbs steeply while average capacity $\bar{m}$ falls as small pods displace cartridges; the nicotine flow into N_{im} rises.

2018–2019 (JUUL, and one caveat for p_{ms}). JUUL becomes the largest brand by mL; representative concentration peaks (~ 34 mg/mL, 2019–20). Two notes for the inputs: (i) the fall-2019 EVALI scare temporarily depressed e-cig demand; and (ii) a **spurious mid-2019 bump** in the *pooled* p_{ms} and nicotine is a two-month liquidation of discontinued 30 mL open-refill bottles (Mistic/HAUS, Cosmic Fog) whose recorded price fell from $\sim \$25$ to $\sim \$2.69$ (verified in the raw Nielsen price field). Because those bottles are large in mL, the mL-weighted $\bar{m}$ jumps ($\sim 3.7 \rightarrow 5.2$), lifting p_{ms} **even though price per mL and the average unit price both fell**. It is real in the data but a *composition/weighting* effect; it is state-specific and small, but it enters the state-month p_{ms} where open refill had share.

2020 (regulation splits the market). The Feb 2020 FDA enforcement priority on flavored *cartridge* ENDS exempted disposables; flavored volume migrates to disposables; pod nicotine plateaus (see the timeline in §4).

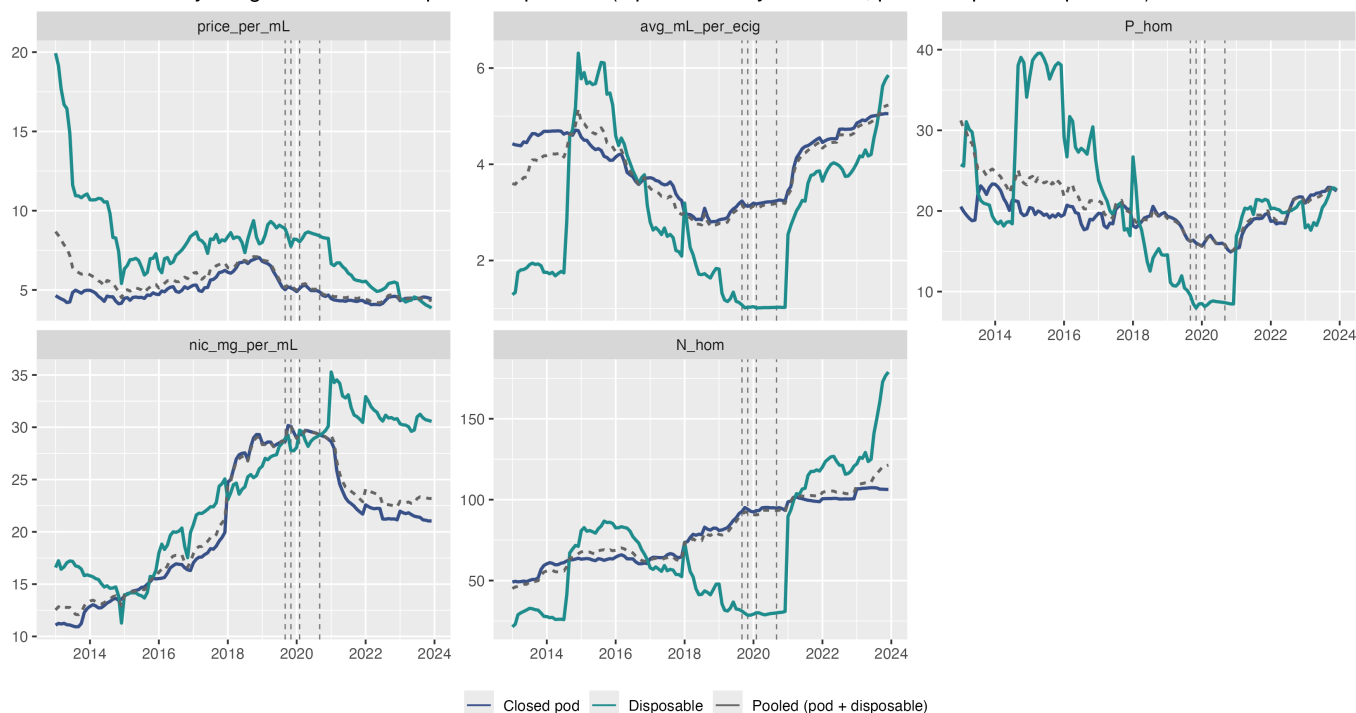
2021–2023 (modern disposables). Within closed pods, price is stable and nicotine keeps rising, now more via capacity than concentration; within disposables, nicotine surges via *size* (5–6 mL units). But see §6 — disposables are a minority of the tracked volume, so p_{ms} remains pod-dominated.

The pooled endpoint decomposition (2013→2023) is robust to all data corrections: $\Delta p_{\text{hom}} \approx -\5.7 (price/mL -13.7 , size $+15.4$) and $\Delta N_{\text{hom}} \approx +80$ mg (concentration $+37$, size $+23$).

3. The same series excluding open refill

The pooled series in §2 include all three product types, and open refill is the noisiest of them: it is a thin sliver of tracked volume (bottled e-liquid and open tanks, sold largely through channels the scanner panel captures poorly), so a single clearance event or a few large-bottle SKUs can swing its mL-weighted averages — the mid-2019 artifact is exactly such an episode. Because the firm-relevant "closed" products are the closed pods and disposables, it is useful to view the series with open refill fully excluded. In this view the pooled line is **pod + disposable only**.

National monthly e-cig series - Closed pod & Disposable (Open refill fully excluded; pooled = pod + disposable)



Events: 1 EVALI scare (fall 2019) 2 JUUL pulls fruit/mint (Nov 2019)
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A note on brand composition first. In the *all-types* market (§8 concentration table), MISTIC is the market leader in 2015–2017 (~21–26% of all mL). But MISTIC is ~75–88% **open refill** — bottled juice — so it is largely absent from the pod+disposable series shown here. Excluding open refill, the closed-product leaders through 2017 are the cigalike incumbents: BLU (61%→26% of pod+disposable mL, 2013→2017), then LOGIC rising to second (14→20%), with 21st Century Smoke, FINITI, and NJOY filling in. This is why the closed-product narrative names Logic rather than MISTIC — MISTIC's mid-decade "leadership" is a bottled-juice phenomenon that the supply-relevant closed series does not see.

Period by period, for the pod+disposable pooled series (P_{hom} in real \$, N_{hom} in mg):

- **2013–2015 (cigalike closed pods + old disposables).** BLU-led cigalikes dominate; price/mL falls from ~\$7.0 to ~\$5.1, nicotine is modest (~13→15 mg/mL), capacity $\bar{m}$ is ~3.9–4.7 mL. P_{hom} eases ~27→24, N_{hom} rises ~49→68.
- **2016–2017 (salts begin, pods shrink).** LOGIC climbs to second; nicotine concentration rises (14.6→19.4 mg/mL) while capacity falls (4.7→3.3 mL) as small pods displace big cartridges. P_{hom} ~24→20; N_{hom} dips slightly (~68→64) because capacity falls a touch faster than concentration climbs.
- **2018–2019 (JUUL).** JUUL takes 51% then 44% of pod+disposable mL; concentration jumps (19.4→28.9 mg/mL) and capacity collapses to ~2.8–3.0 mL (0.7 mL JUUL pods); N_{hom} 64→86. Crucially, with open refill excluded there is **no mid-2019 bump** — the spurious pooled-series blip in §2 was entirely an open-refill clearance artifact, and it vanishes here. This is the clearest demonstration of why the no-open view matters.
- **2020 (regulation splits the market).** JUUL still leads (34%) but VUSE closes to 28%; nicotine plateaus (~29 mg/mL), capacity begins rising again; N_{hom} ~93. See the timeline in §4.
- **2021–2023 (VUSE/JUUL duopoly, modern disposables).** VUSE leads (44%→55%), JUUL second; capacity rises sharply (3.2→5.0 mL) as 5–6 mL modern disposables grow; concentration eases (29→23 mg/mL) but N_{hom} keeps climbing (93→115) because size rises faster than concentration falls; price/mL settles near \$4.4 and P_{hom} recovers ~16→22.

The endpoint decomposition is robust to excluding open refill: 2013→2023, $\Delta P_{\text{hom}} \approx -\5.2 (vs $-\$5.7$ pooled-with-open) and $\Delta N_{\text{hom}} \approx +66$ mg (vs $+80$ pooled-with-open) — the same sign and story, with the smaller N change reflecting that large, high-nicotine open-refill bottles inflate the pooled endpoint.

4. Timeline of regulatory and market events

The following events provide context for the 2019–2020 shifts in the series above; the dashed vertical lines in the figures mark them.

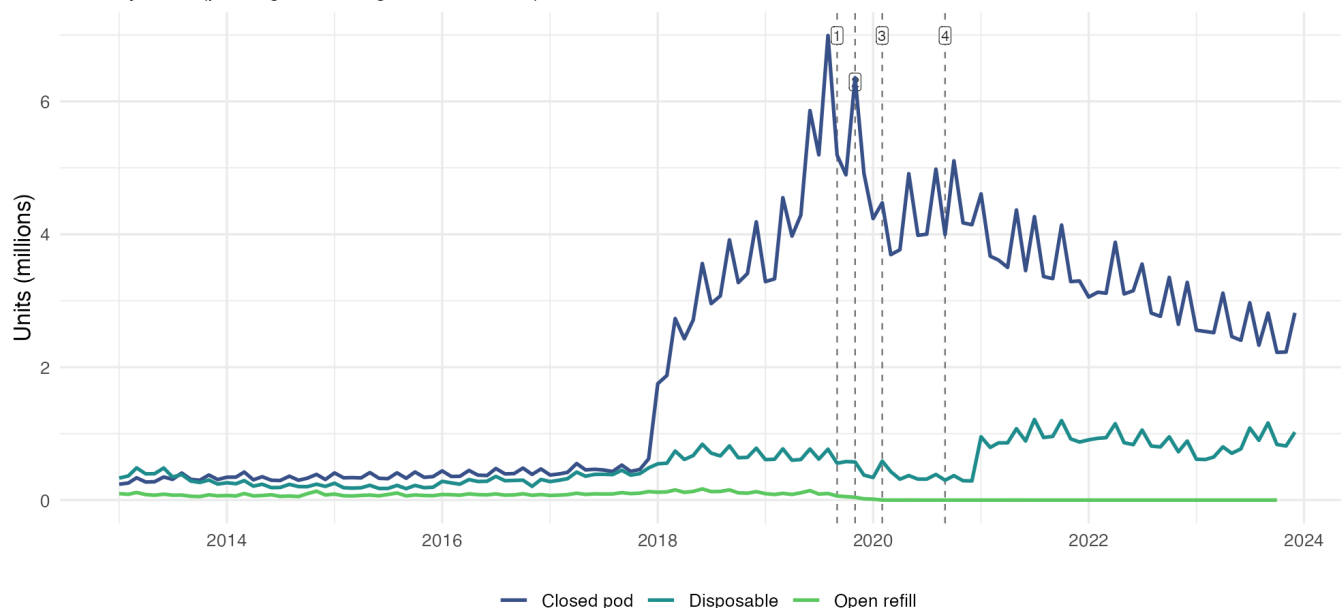
- Fall 2019: the EVALI outbreak (peaked August–September 2019), linked mostly to illicit THC/vitamin-E-acetate products rather than nicotine pods, temporarily depressed nicotine e-cigarette demand.
- September 2019: the administration announced its intent to clear flavored e-cigarettes from the market; JUUL voluntarily pulled its fruit and mint flavors in November 2019.
- January 2, 2020: the FDA finalized its "Enforcement Priorities for ENDS" guidance.
- February 6, 2020: the enforcement priorities took effect, prioritizing action against flavored (other than tobacco and menthol) prefilled cartridge / pod ENDS. Open-system e-liquids and disposables were not covered, so flavored disposables (Puff Bar, later Elf Bar) were exempt and flavored volume migrated to them.
- September 9, 2020: the premarket tobacco application (PMTA) deadline — products had to have submitted applications to remain legally marketed. Many modern disposables are imports without granted PMTAs, which is part of why they sell through channels the scanner panel captures poorly.

5. Tracked sales volume (units)

The trends above work in shares and in price/nicotine intensities; the figures below give the underlying denominator — total tracked unit sales — so the shares can be read in context. Units are packages/cartridges as scanned (see §7, the unit-definition caveat, regarding cross-source comparisons).

Tracked e-cigarette unit sales by product type (RMS)

Monthly units (packages/cartridges as scanned), millions

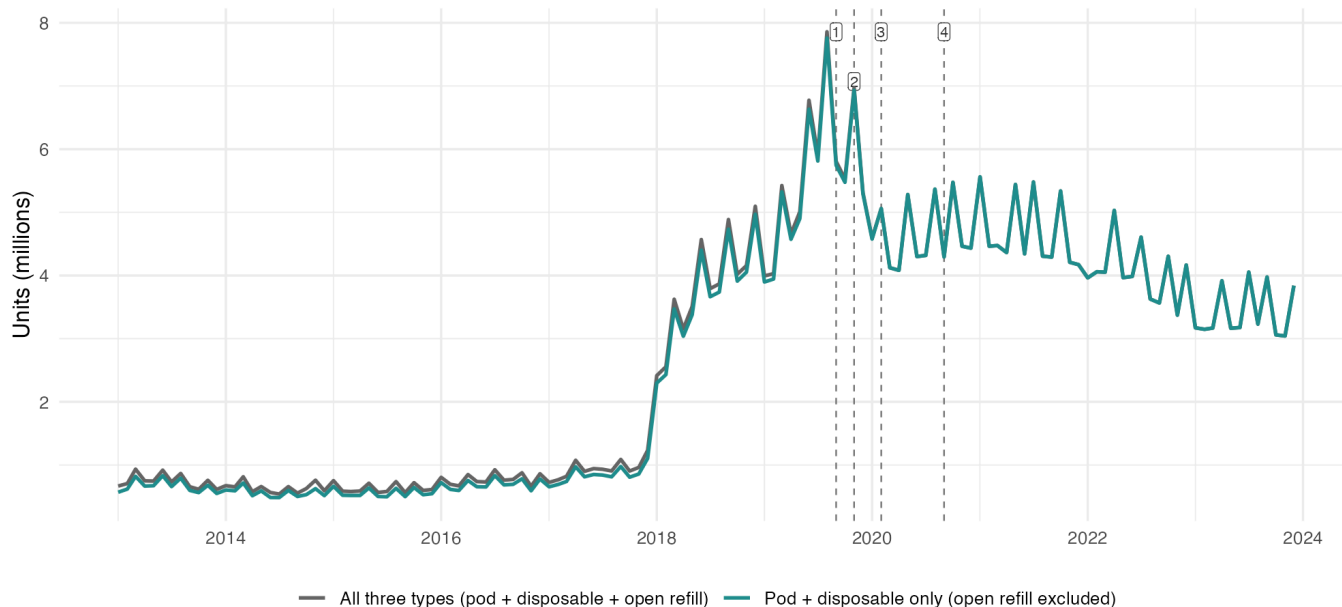


Events: 1 EVALI scare (fall 2019) 2 JUUL pulls fruit/mint (Nov 2019)
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Two features stand out: the ~2018 step up in closed-pod units (the JUUL takeoff), and the post-2020 re-emergence of disposables. Open refill is small throughout and falls to near zero after 2020.

Tracked e-cigarette unit sales, pooled (RMS)

Open refill is a small share of tracked units; the two lines nearly coincide after 2019



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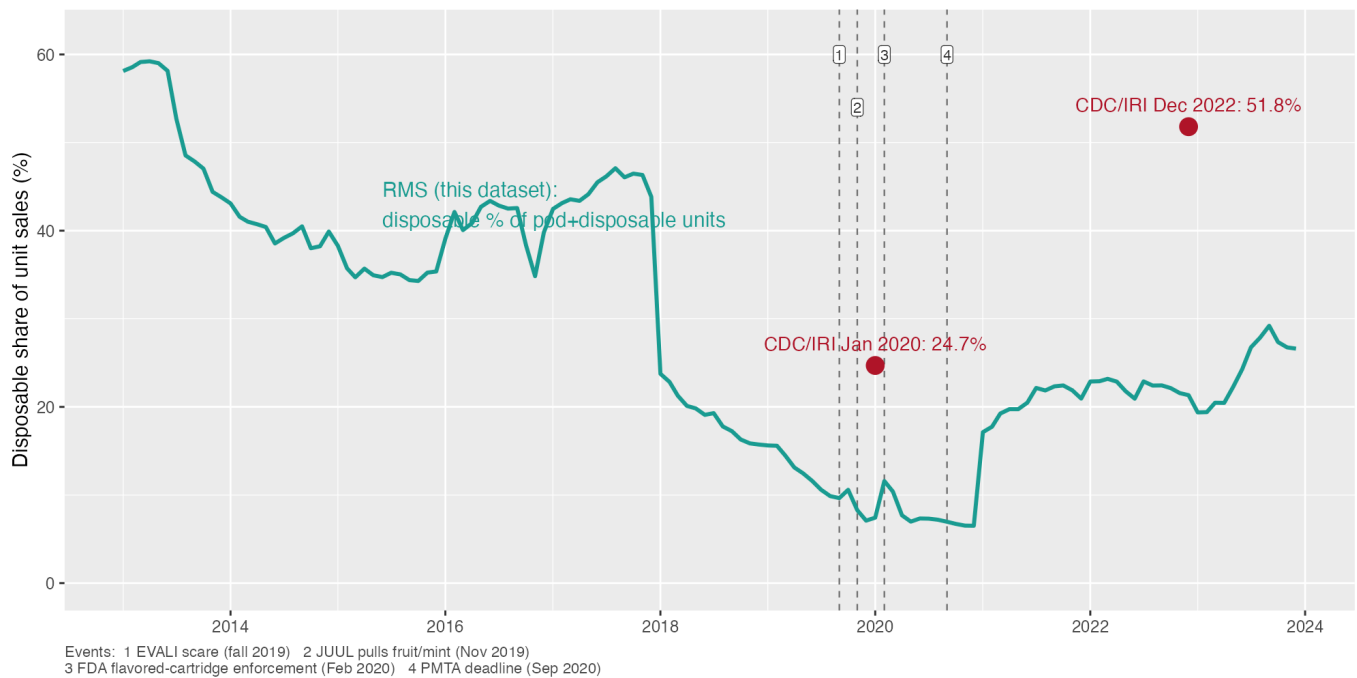
Pooling confirms that open refill contributes little to tracked *unit* volume: the all-three-types line and the pod+disposable-only line nearly coincide, especially after 2019. This is the volume counterpart to the mL-weighting point in §6 — the tracked market is overwhelmingly a closed-product (pod, then pod+disposable) market by units.

6. Disposables are under-represented in RMS — implication for p_{ms}

The representative p_{ms} is mL-weighted and therefore pod-dominated, partly because disposables are a small share of *tracked* volume: ~24% of units and ~17% of mL even in 2023. Benchmarking against the CDC/IRI all-outlet series shows the gap.

Disposable share of e-cigarette unit sales: RMS vs CDC/IRI all-outlet benchmark

RMS tracks the same rise but captures only ~30-40% of the all-outlet disposable share



: Disposable share of unit sales (disposable / (pod + disposable)).

	Jan 2020	Dec 2022
CDC/IRI (all-outlet)	24.7%	51.8%
RMS (this dataset)	7.4%	21.3%

The CDC MMWR (IRI retail-scanner data) reports the disposable unit share rising from **24.7% to 51.8%** over Jan-2020→Dec-2022 — disposables became the market majority — while the RMS series shows the same *direction and timing* at ~30–40% of the *level*. The gap is a post-2020, **modern-disposable** phenomenon: those brands (Puff Bar, Elf Bar, Breeze, Lost Mary) sell heavily through vape shops, specialty/convenience, and online, and are largely **imports without granted PMTAs** — sales channels scanner panels capture poorly (and even IRI excludes vape shops/online).

For the paper this is a scope statement, not a flaw. The representative e-cigarette is the *tracked-retail, pod-weighted* e-cigarette, and the household panel purchases come from the same tracked universe, so demand and the price index are internally consistent. Two points are worth stating: (i) that the representative e-cig is pod-weighted and disposables are a lower bound; and (ii) that this connects to the "Moving Forward" idea of capturing the changing e-cig landscape (number of brands/products) — the disposable wave is exactly the heterogeneity RMS misses.

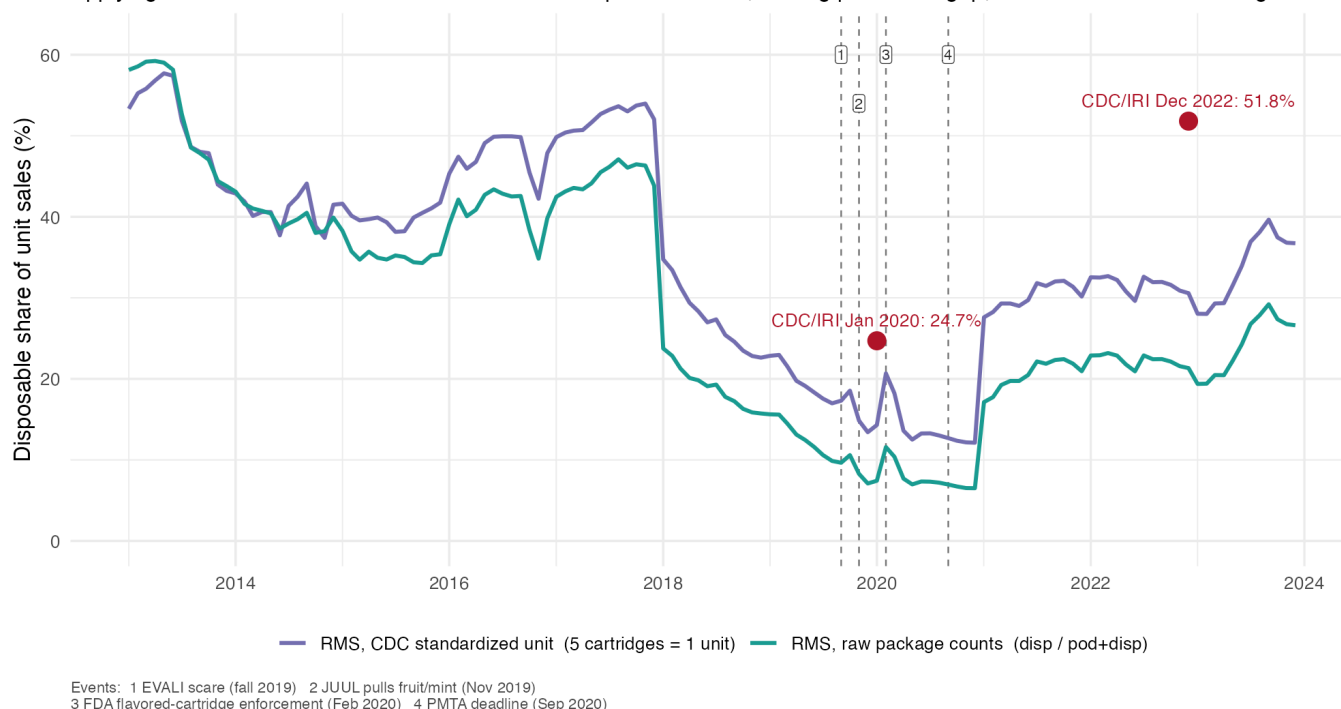
7. The low RMS disposable share: unit-definition vs. channel coverage

The RMS disposable share in §6 sits well below the CDC/IRI benchmark. Two things could drive that: (i) a *unit-definition* difference — the CDC/IRI series standardizes **1 unit = 5 prefilled cartridges = 1 disposable device = 1 e-liquid bottle** (Ali et al. 2023), whereas the §6 RMS series counts raw scanner packages, so a multi-pod pack counts as one "unit" against one disposable; and (ii) genuine *channel* undercoverage. The figure isolates (i) by recomputing the RMS disposable share under CDC's standardized unit — closed-pod cartridges ÷ 5, one unit per disposable and per e-liquid bottle — and

including e-liquid in the denominator to match CDC's three categories. This standardization is a comparability diagnostic only; it is **not** used in the demand or supply model, which keep the raw series.

Does RMS understate the disposable share? Unit-definition diagnostic

Applying CDC's standardized unit raises the RMS disposable share, closing part of the gap; the rest is channel coverage



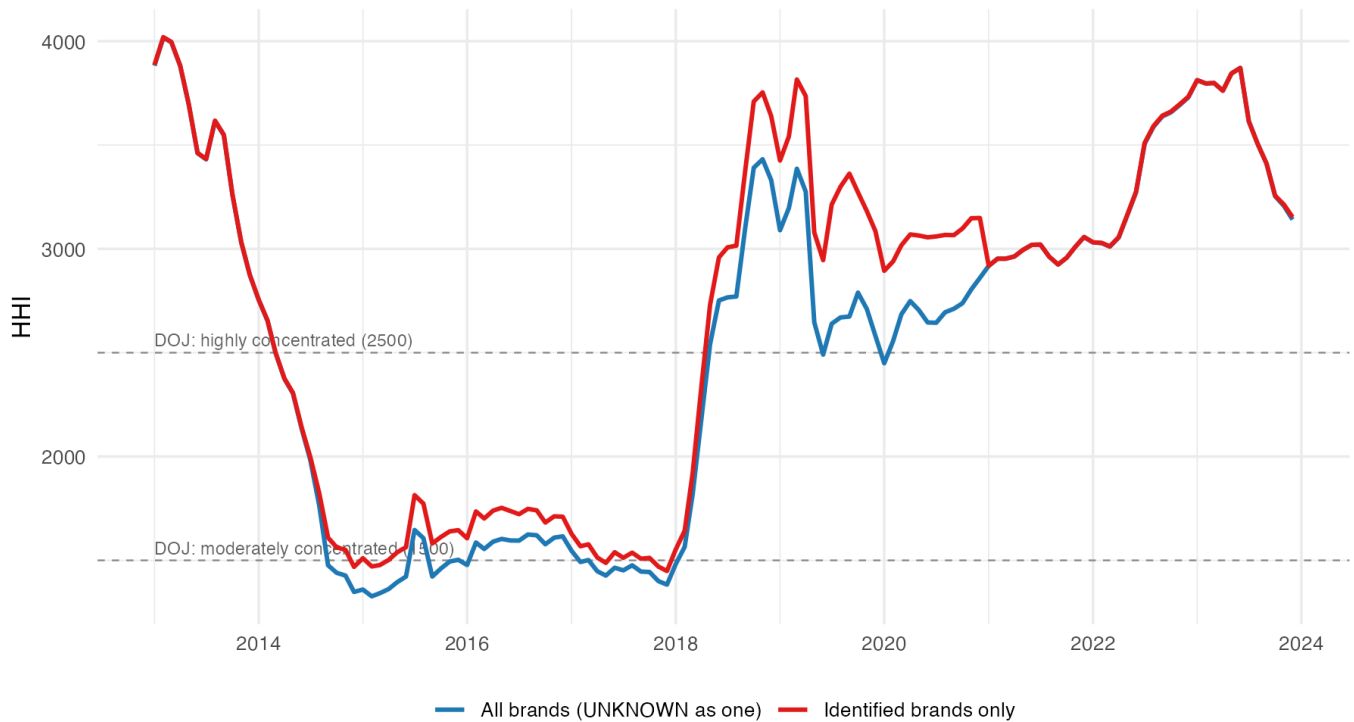
The standardized definition raises the RMS disposable share materially — from 7.4% to 14.3% in January 2020, and from 21.3% to 30.6% in December 2022 — closing roughly 30–40% of the gap to the CDC/IRI points (24.7% and 51.8%). The residual gap (14.3 vs 24.7; 30.6 vs 51.8) is what channel coverage explains: IRI captures convenience/gas/dollar/ military outlets that carry disposables heavily, and both IRI and RMS exclude vape shops and online, where modern disposables concentrate. So the low RMS disposable share reflects two compounding causes — the unit convention and outlet coverage — and even the standardized RMS share remains a lower bound on the true all-channel figure. (Cartridge counts here are recovered from the always-populated pack structure; disposables and bottles are one unit each.)

8. Market concentration over time — and what it implies for conduct

The homogeneous-good series average over whichever brands sell in a given state-month, so the concentration of that underlying brand set is itself a supply-relevant object. The figure plots the monthly Herfindahl–Hirschman index (HHI) of national brand mL shares — mL-weighted to match $P_{\text{hom}}/N_{\text{hom}}$ — on the 0–10,000 scale with the DOJ moderately-/highly-concentrated reference lines (1,500 and 2,500). Two series are shown: all brands (the residual **UNKNOWN** bucket treated as one "brand," which understates fragmentation) and identified brands only (UNKNOWN dropped, shares renormalized).

National e-cigarette market concentration (HHI, mL-weighted)

Monthly Herfindahl index of brand mL shares, 0-10,000 scale

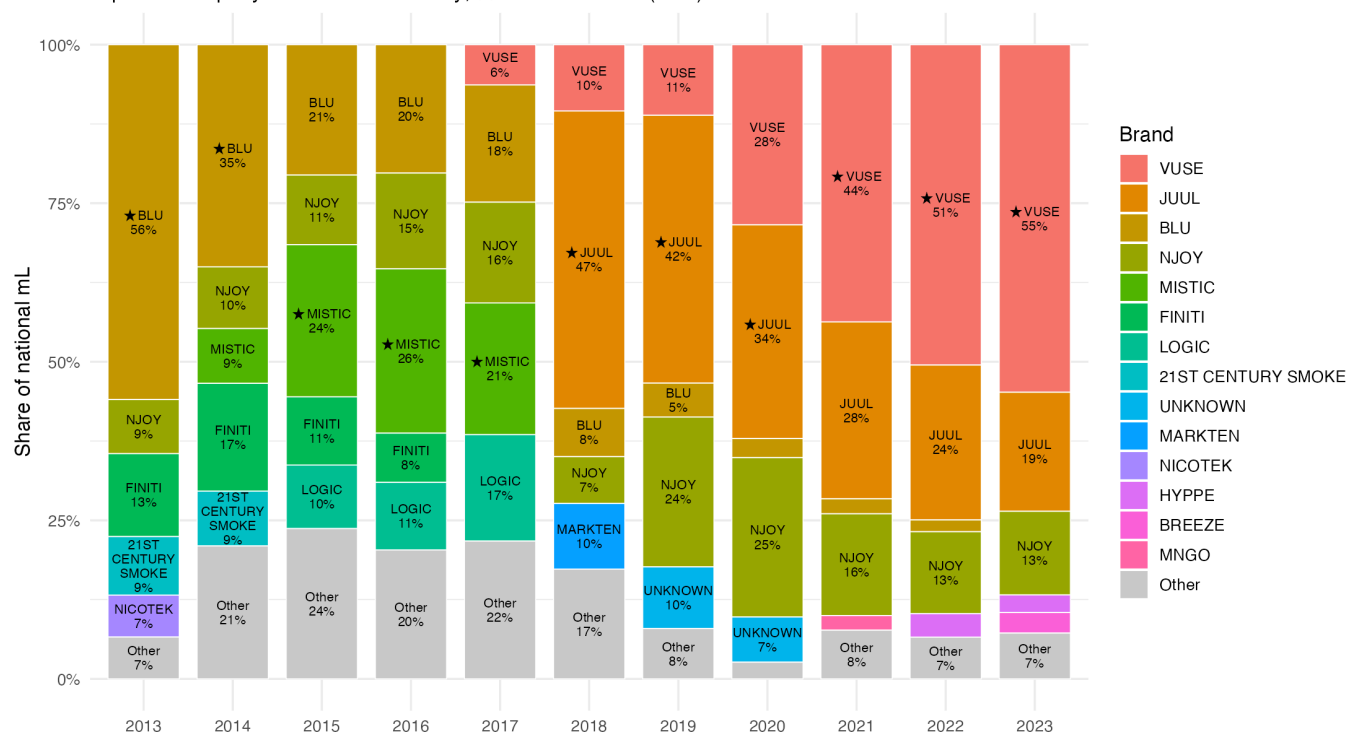


The concentration path is **U-shaped**:

- **2013 (BLU era).** BLU holds ~56% of tracked mL; HHI ~3,500–4,000 — highly concentrated.
- **2014–2017 (fragmentation).** MISTIC/NJOY/LOGIC and bottled-juice brands proliferate; no brand exceeds ~26% and HHI falls to ~1,400–1,600, near the moderately-concentrated floor.
- **2018–2020 (JUUL).** JUUL re-concentrates the market; HHI back above 2,500, leader share ~47% (2018).
- **2021–2023 (VUSE/JUUL duopoly).** VUSE overtakes JUUL; the top two hold ~88% of tracked mL (CR3 ≈ 87%) and HHI climbs to ~3,550 by 2023 — highly concentrated again.

Annual e-cigarette brand composition (mL-weighted)

Top-5 brands per year shown individually; ★ = market leader (CR1). Rest = "Other".



: Annual concentration, mL-weighted brand shares (national).

Year	Leader	Leader share	Runner-up	CR3	HHI
2013	BLU	56.0%	FINITI	78.3%	3519
2014	BLU	35.0%	FINITI	61.8%	1869
2015	MISTIC	24.0%	BLU	55.5%	1430
2016	MISTIC	25.9%	BLU	61.3%	1564
2017	MISTIC	20.8%	BLU	56.0%	1430
2018	JUUL	46.9%	VUSE	67.7%	2612
2019	JUUL	42.3%	NJOY	77.0%	2609
2020	JUUL	33.7%	VUSE	87.3%	2639
2021	VUSE	43.7%	JUUL	87.7%	2966
2022	VUSE	50.5%	JUUL	87.9%	3338
2023	VUSE	54.8%	JUUL	86.8%	3553

Implication for the supply side (structure vs. conduct). For most of the sample the tracked market is *highly* concentrated — a single dominant brand (BLU, then JUUL) and, by 2021–2023, a VUSE/JUUL duopoly with HHI ~3,500. Taken at face value, that structure is **not** obviously a price-taking, competitive market, so the competitive / constant-marginal-cost closure recommended in §9 cannot be justified on *structural* grounds. That is precisely why the recommendation leans on **observed near-unit tax pass-through** rather than on the HHI: pass-through is firm *behavior*, and — given the demand curvature the logit implies — full pass-through of a *specific* tax maps to $\theta \approx 0$, whereas a structural HHI is silent about

conduct. Two takeaways: (i) the concentration series is the reason to *test* conduct through pass-through rather than *assume* competition; and (ii) because structure points toward market power, imposing $\theta = 0$ is a genuine assumption, worth a robustness check that either bounds θ or re-runs the ban counterfactual under a positive constant markup (§9(e) already sketches why a positive markup would make the "modest increase" an upper bound).

One data caveat mirrors §6: this HHI is computed on **tracked-retail mL**, where modern disposables (Puff Bar, Elf Bar, Breeze, Lost Mary) are undercovered. Because those brands sell through channels the scanner panel captures poorly, the true post-2020 market is likely *less* concentrated than the pod-dominated series shows — the 2021–2023 HHI is an upper bound on all-channels concentration, even as it accurately describes the tracked-retail universe the demand model uses.

9. A modest supply side that fits the model

Purpose. A 40-page supply model is not needed. A one- to two-equation supply side (a) *closes* the model so counterfactual **prices can respond**, (b) is **disciplined by observed tax pass-through**, and (c) demonstrates the e-cigarette ban result is **robust** to price responses. Three design choices follow from the setup.

(a) Match the product; avoid BLP. The demand model puts the e-cigarette in as one homogeneous representative good at the state-month level. The supply side should be a **homogeneous-good pricing relation at the state-month level**. A differentiated products (BLP) pricing system over pods/disposables/brands is inconsistent with a homogeneous demand object and is unnecessary.

(b) Framework: a conduct (markup) pricing relation. The natural counterpart is the conduct-parameter / conjectural-variations relation (Bresnahan 1982; Lau 1982) — the homogeneous-good tool within the broader New Empirical IO program (to be precise: NEIO is the research program; the conduct relation is the specific model). For the representative e-cigarette with own-price elasticity η_{ms} implied by the demand estimates,

$$p_{ms} = MC_{ms}(w_{ms}, \tau_{ms}, N_{ms}) - \theta \frac{p_{ms}}{\eta_{ms}},$$

where τ_{ms} is the excise tax, w_{ms} input-cost shifters, N_{ms} the representative nicotine content, and $\theta \in [0, 1]$ nests competition ($\theta = 0, p = MC$) through monopoly ($\theta = 1$).

(c) Recommendation: impose the competitive / constant-marginal-cost case rather than estimate θ . Two reasons, both grounded in the paper's own materials:

- Cotti et al. (2022) put e-cig tax pass-through at 0.90–1.01 and He et al. (2023) put cigarette pass-through at 1.00–1.10. Full pass-through of a specific tax is the signature of competitive / constant-MC pricing; it is hard to reconcile with substantial market power plus the demand curvature the logit implies. So the data point to $\theta \approx 0$.
- *Identification.* Separating conduct θ from marginal cost requires **demand rotators** — instruments that change the *slope* of demand. The only excluded price instrument is the excise tax, which is *already* used to instrument price in the demand second stage; a tax shifts both cost and (through price) demand level, so it cannot also credibly identify conduct (cf. Corts 1999). Imposing the competitive/full-pass-through case is therefore both more defensible and more parsimonious than estimating a weakly-identified θ .

Concretely: set $p_{ms} = c_{ms} + \tau_{ms}$ (or $c_{ms} + \tau_{ms} + \text{a constant markup}$), with c_{ms} a marginal-cost function of input prices and the representative nicotine content N_{ms} ; back out c_{ms} from observed prices and taxes; validate against the published pass-through. This adds essentially one equation.

(d) Nicotine belongs in cost, not a firm FOC. In the model nicotine enters *dynamically* through the addiction stock N_{im} , and the representative nicotine is a mL-weighted market aggregate rather than a single firm's static choice variable. Nicotine should therefore enter **marginal cost** (higher delivered nicotine is costlier and, post-2016, technologically gated by salts) and be treated as a slowly evolving characteristic — *not* a structural nicotine first-order condition. A firm-level nicotine-choice model would require a differentiated-products setup that has been deliberately avoided.

(e) Why this is worth a paragraph in the paper — the ban counterfactual. The e-cigarette ban raises cigarette demand by removing a substitute. With **fixed prices** (the current counterfactual) the result is a *modest* increase in cigarette smoking. Adding the competitive/constant-MC supply side leaves cigarette marginal cost — and hence the cigarette price — **unchanged** when e-cigarettes are banned, so the fixed-price ban result is *robust* under the pricing model the pass-through evidence supports. The contrast is informative: *if* cigarette firms had market power, the demand increase would raise cigarette prices and **attenuate** the switch to cigarettes, making the "modest increase" an upper bound. Near-unit cigarette pass-through (He et al. 2023) argues against that, so the competitive closure both validates the result and tells the reader why. For **tax** counterfactuals the same supply relation delivers the price response directly.

10. Summary and what would sharpen this

- The p_{ms} and representative nicotine are mL-weighted homogeneous objects; the pooled series carry one small mid-2019 open-refill artifact worth a footnote, and are pod-weighted because disposables are undercovered in tracked retail (a scope statement, consistent with the household panel).
- Tracked-market concentration is **U-shaped** (HHI $\sim 3,500$ under BLU in 2013, down to $\sim 1,400$ – $1,600$ in 2015–2017, back to $\sim 3,500$ under the VUSE/JUUL duopoly by 2023). Because the market is highly concentrated for most of the sample, the competitive closure rests on *observed pass-through*, not structure — and imposing $\theta = 0$ is an assumption worth a robustness check.
- Pair the homogeneous-good demand with a **homogeneous-good, state-month pricing relation; impose competitive / constant-MC with full tax pass-through** (justified by Cotti et al. 2022 and He et al. 2023), rather than estimate a weakly-identified conduct parameter; put nicotine in marginal cost. Avoid BLP and avoid a demand/supply product mismatch (no pod-only supply under all-e-cig demand).
- The supply side's payoff is concrete and modest: it disciplines price/tax counterfactuals and demonstrates the e-cigarette-ban result is robust to price responses.

Making the supply moments exact requires: (i) the implied own-price elasticity of the representative e-cigarette from the nested-logit estimates (given $\hat{\beta}^p$, $\hat{\lambda} = 0.624$, and the shares); (ii) a decision on whether the supply side is wanted purely for *counterfactual price response* or also to *estimate* marginal cost; and (iii) whether cigarettes and e-cigarettes should share a cost/pass-through specification or be separate. With those, the exact estimating/closing equations can be written to slot into the GMM.